Jingyuan Hu

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Academic Appointments and Research Training

Harvard T.H. Chan School of Public Health Boston, MA
Postdoctoral Research Fellow January 2025–present
Advisor: Rong Ma, Ph.D.

Dana-Farber Cancer Institute Boston, MA
Postdoctoral Research Fellow January 2023–December 2024
Advisor: Sahand Hormoz, Ph.D.

Education

Baylor College of Medicine Houston, TX
Ph.D., Quantitative and Computational Biosciences December 2022
Dissertation: "A Computational Framework for CRISPR-Based Single-Cell Lineage Tracing"
Advisor: Zhandong Liu, Ph.D. July 2017–December 2022

The George Washington University Washington, DC
B.S., Biological Sciences December 2016
Advisors: Jeremy Goecks, Ph.D., and Keith Crandall, Ph.D. July 2014–December 2016

Harbin Institute of Technology Harbin, China
Coursework toward a major in Electronic Engineering September 2012–July 2014

Publications

1. **Hu, J.**, Singh, H., Das, J., Pinello, L., & Ma, R. (2026). FlowMap: Geometry-preserving embedding of RNA velocity for interpretable cellular dynamics. *bioRxiv preprint, forthcoming.*
2. Ma, R.*, Li, X.*, **Hu, J.**, & Yu, B. (2026). Uncovering smooth structures in single-cell data with neighbor embeddings guided by predictability-computability-stability. *Patterns*, 101621. doi:10.1016/j.patter.2026.101621.
3. Singh, S., Lee, N., Pedroza, D. A., Bado, I. L., Hamor, C., Zhang, L., Aguirre, S., **Hu, J.**, Shen, Y., Xu, Y., & Gao, Y. (2022). Chemotherapy coupled to macrophage inhibition induces T-cell and B-cell infiltration and durable regression in triple-negative breast cancer. *Cancer Research*, *82*(12), 2281–2297.
4. Gong, W.*, Grandos, A.*, **Hu, J.***, Jones, M.*, Raz, O., Salvador-Martinez, I.*, Zhang, H.*, ... & Meyer, P. (2021). Benchmarked approaches for cell lineage reconstructions from in vitro dividing cells and in silico models of *Caenorhabditis elegans* and *Mus musculus* development. *Cell Systems.*
5. Bado, I. L.*, Zhang, W.*, **Hu, J.**, Xu, Z., Wang, H., Sarkar, P., ... & Zhang, X. H. F. (2021). The bone microenvironment increases phenotypic plasticity of ER+ breast cancer cells. *Developmental Cell.*
6. Zhang, W.*, Bado, I. L.*, **Hu, J.**, Wan, Y. W., Wu, L., Wang, H., ... & Zhang, X. H. F. (2021). The bone microenvironment invigorates metastatic seeds for further dissemination. *Cell.*

**Equal contribution.*

Honors and Awards

- Winner, Allen Institute Lineage Reconstruction DREAM Challenge, 2020.

Invited Talks and Presentations

- "FlowMap: Geometry-Preserving Embedding of RNA Velocity." Models, Inference & Algorithms (MIA), 2026.
- Invited talk, Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS), Shanghai, China, 2026.
- "Lineage Reconstruction Algorithm for the Allen Institute Lineage Reconstruction DREAM Challenge." DREAM Challenge Consortium, 2020.
- "Lineage Reconstruction Algorithms for Single-Cell Genomics Data." Graduate Student Annual Symposium, Baylor College of Medicine, 2020.
- "Lineage Reconstruction Algorithms for Single-Cell Genomics Data." Rice University, 2020.

Conference Presentations and Posters

- Poster presentation (upcoming), New England Computational Biology Symposium, Cambridge, MA, October 2026.
- Poster presentation (upcoming), Impact of Genomic Variation on Function (IGVF) Consortium Annual Meeting, 2026.
- "FlowMap: Geometry-Preserving Embedding of RNA Velocity." Harvard T.H. Chan School of Public Health Postdoctoral Association Poster Day, Boston, MA, April 2026.
- "CRISPR Lineage Tracing." Graduate Student Annual Symposium, Baylor College of Medicine, Houston, TX, 2021.

Software

- **FlowMap** (documentation): Python package for geometry-aware vector field embeddings of RNA velocity and other dynamical single-cell data.
- **Iso-Seq Browser**: Interactive genome browser for Pacific Biosciences long-read isoform sequencing data.

Professional Service

Journal peer review: *Cell Reports*; *Frontiers in Genetics*; *Journal of Machine Learning Research* (2026); *Scientific Reports* (2026); *Journal of Complex Networks* (2025–2026).
Conference review: ISMB 2026 (two submissions).